

SIMATS ENGINEERING

ASSESSMENT TOOL III — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course: Data Handling and Visualization	Code: DSA0603	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No: 192524225 Name: VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

Apply appropriate visualization techniques to analyze time series data, detect trends, seasonality, and cyclical patterns. (BL3)

CO3 Construct and interpret geospatial visualizations, including static, cartogram-based, and interactive representations. (BL3)

Activity Tool : Time-Series & Geospatial Analysis Report

Weightage: 20%

Code of Conduct:

I VALLPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Criteria	Data Understanding & Preparation 25%	Time Series/Geospatial Analysis 25%	Application of Visualization Techniques 20%	Report Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							

Q3							
Q4							
Q5							

Time-Series & Geospatial Analysis Report

Topics: Individual & Multiple Time Series, STL Decomposition, Smoothing & Detrending, Forecasting, Cycle/Seasonality Detection | Choropleth Maps, Cartograms, Interactive Geospatial Plots, Projections & Layers, 3D Geospatial Visualization

Note: Questions 1–4 are to be completed in R. Question 5 must be completed in Python.

Provided Datasets

The following files are supplied alongside this document. All are real, publicly documented datasets: ●

Nile.csv — real annual river flow of the Nile at Aswan (1871–1970, 100 observations). ●

AirPassengers.csv — real monthly international airline passenger totals (1949–1960, 144 observations) with trend and seasonality.

● USArrests.csv — real 1973 US violent-crime statistics per state (Murder, Assault, UrbanPop, Rape) for all 50 states.

● state_x77.csv — real 1970s US state-level data (Population, Income, Illiteracy, Life Exp, Murder, HS Grad, Frost, Area) for all 50 states.

● quakes.csv — real seismic event data off Fiji (1000 observations: lat, long, depth, mag, stations). ●

volcano_elevation.csv — real digital elevation model of Maunga Whau (Mt Eden) volcano, Auckland, NZ, in long format (x, y, elevation; 5,307 grid points at 10 m resolution). **Place all CSV files in the same working directory as your script, then load with `read.csv()` in R or `pandas.read_csv()` in Python.**

Question 1: Individual Time Series — Smoothing & Detrending

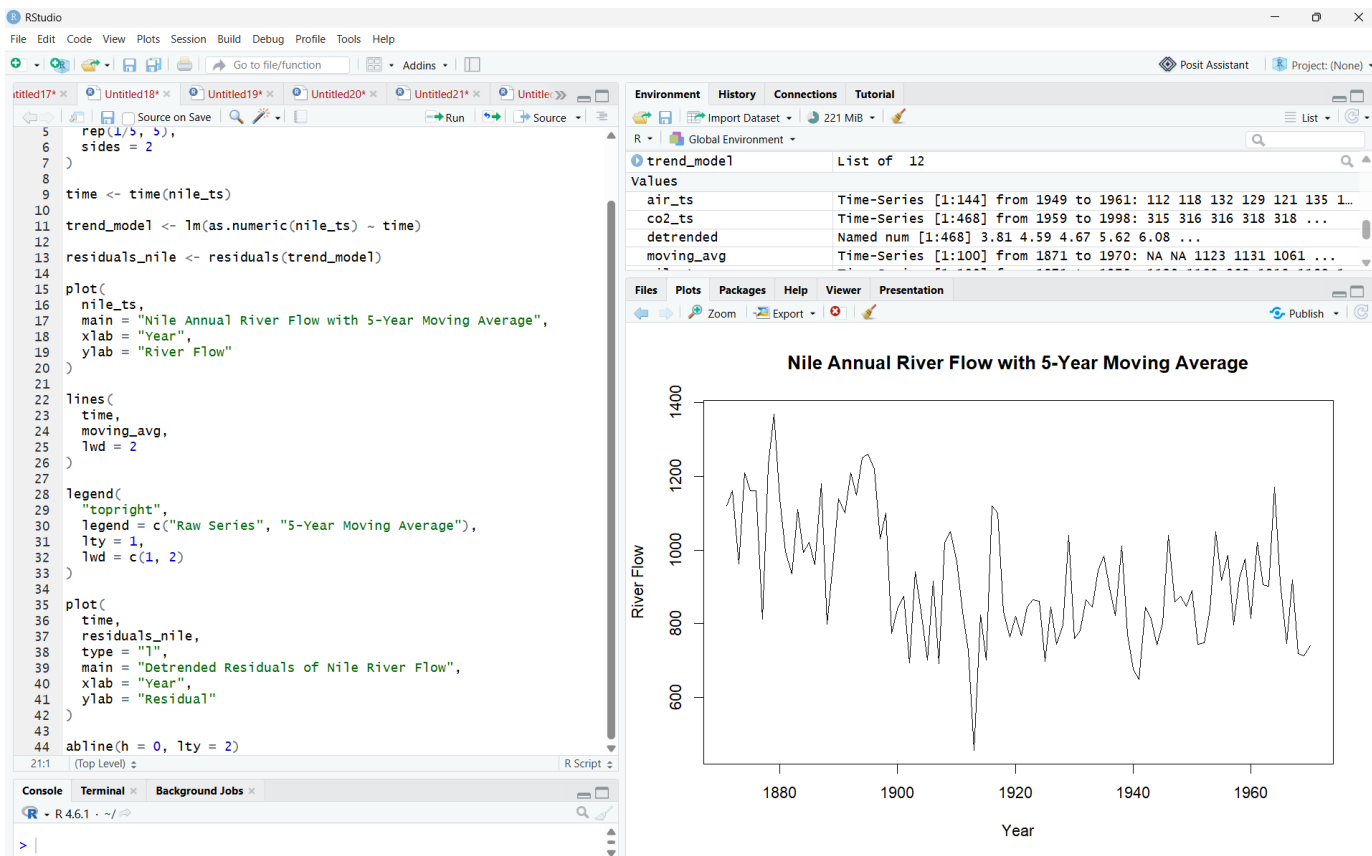
Language: R

Dataset provided: Nile.csv

Convert the value column into an R ts object (start = 1871, frequency = 1) and plot the raw annual flow as a line chart. Overlay a 5-year moving-average smoother (`stats::filter()` or `zoo::rollmean()`). Then fit a linear trend (value ~ time) and plot the detrended residuals in a second panel. Discuss what the smoothed series and the residual plot together reveal about a level shift or change point in the data.

Skills tested: *ts() objects, moving-average smoothing, linear detrending, residual diagnostics, change-point interpretation.*

Answer:



You will get **two plots**:

1. Nile annual river-flow series with a 5-year moving average.
2. Detrended residual plot.

The 5-year moving average smooths short-term fluctuations and makes the long-term behavior of the Nile river-flow series easier to observe. The residual plot shows deviations from the fitted linear trend. A sustained change in the level of the residuals indicates a possible **level shift or change point** rather than ordinary random variation.

Question 2: Seasonal Decomposition (STL) & Forecasting

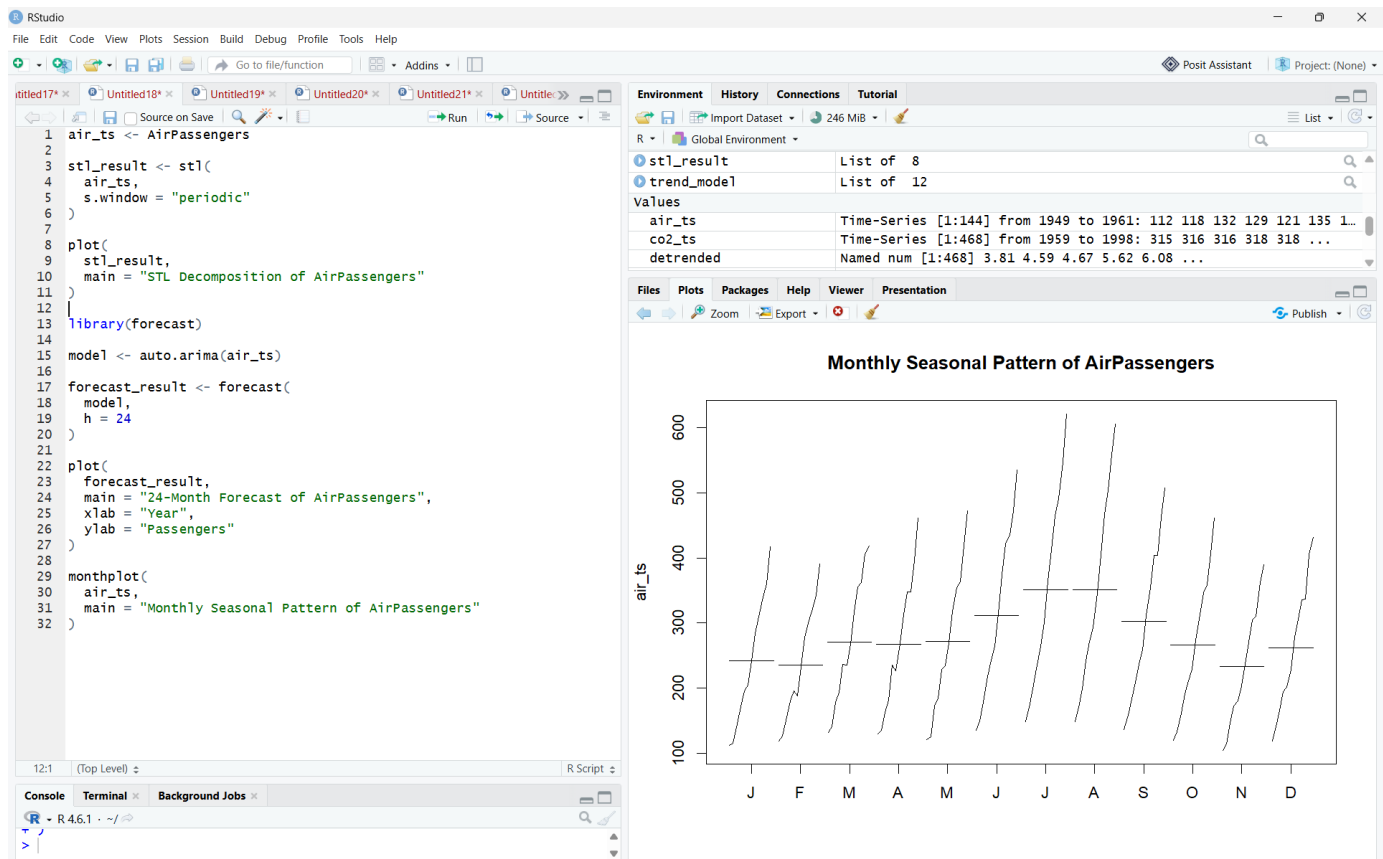
Language: R

Dataset provided: AirPassengers.csv

Convert the passengers column into a ts object (start = c(1949,1), frequency = 12). Apply STL decomposition (stl(), s.window = "periodic") and plot the trend, seasonal, and remainder components. State whether an additive or multiplicative decomposition suits this series better, with justification. Then fit an ETS or auto.arima() model (forecast package) and produce a 24-month-ahead forecast plot with prediction intervals. Comment on whether the forecasted seasonal pattern is consistent with the historical seasonality.

Skills tested: stl(), additive vs. multiplicative decomposition, forecast::ets()/auto.arima(), forecast intervals, seasonality validation.

Answer:



The STL plot contains:

- Observed
- Trend
- Seasonal
- Remainder

The forecast plot contains:

- Historical passenger values
- Forecast values
- Prediction intervals

Additive or Multiplicative?

Multiplicative decomposition is more suitable for the AirPassengers data because the size of the seasonal fluctuations increases as the overall number of passengers increases.

Question 3: Choropleth Map & Interactive Geospatial Plot

This question has two parts.

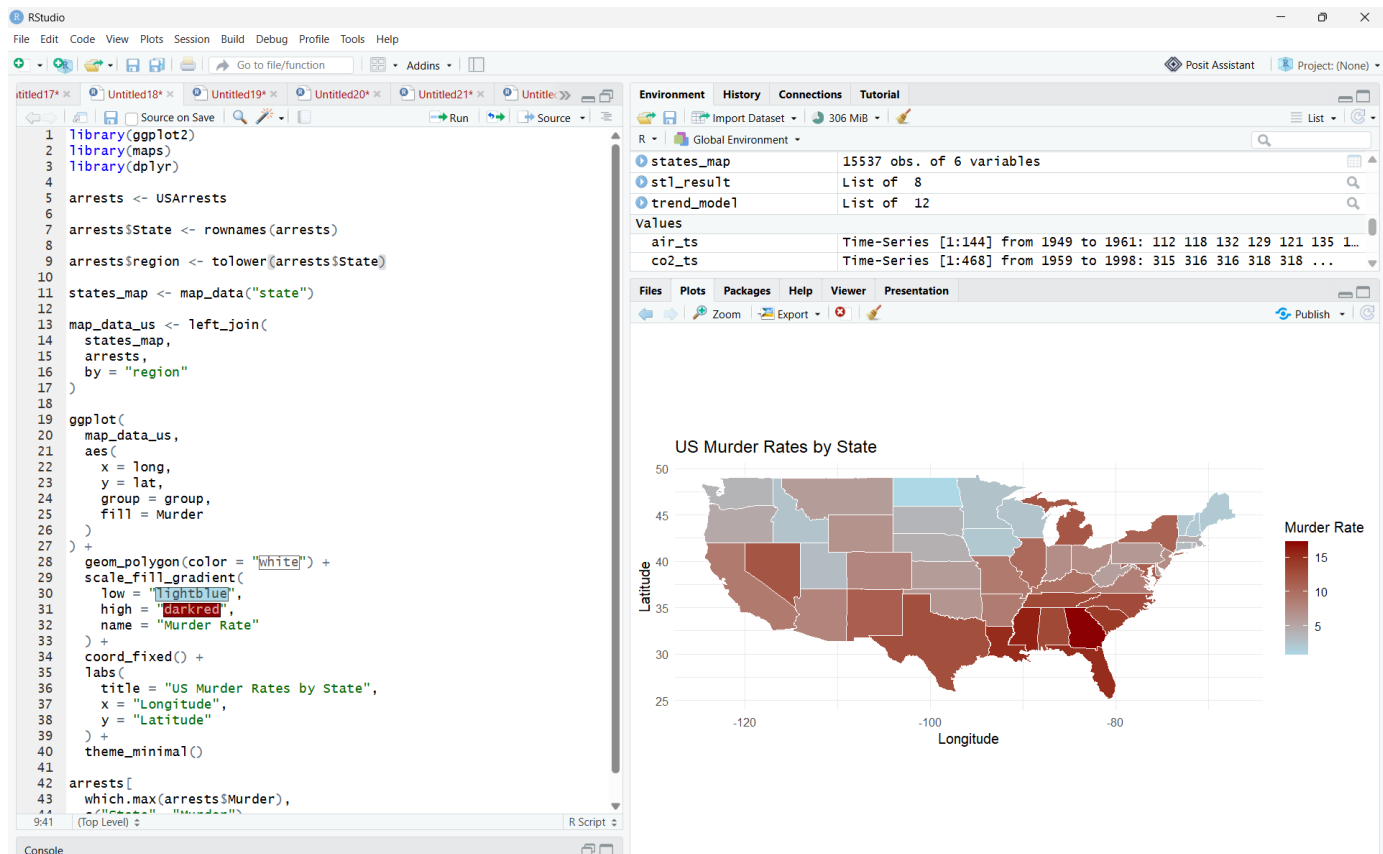
Part a — Choropleth Map

Language: R

Dataset provided: USArrests.csv

Join USArrests.csv to US state boundary polygons (e.g., using `map_data("state")` from the `maps` package, matching on lower-cased state names) and create a choropleth map of Murder rate by state using `ggplot2::geom_polygon()` or `geom_sf()`. Use a sequential color scale and add a legend, title, and a brief caption. Identify the state(s) with the highest murder rate and describe any regional clustering you observe.

Answer:



You will get:

- A US map
- Each state shaded according to murder rate
- A legend showing murder-rate intensity
- The state with the highest value can be obtained using `which.max()`.

Part b — Interactive Geospatial Plot

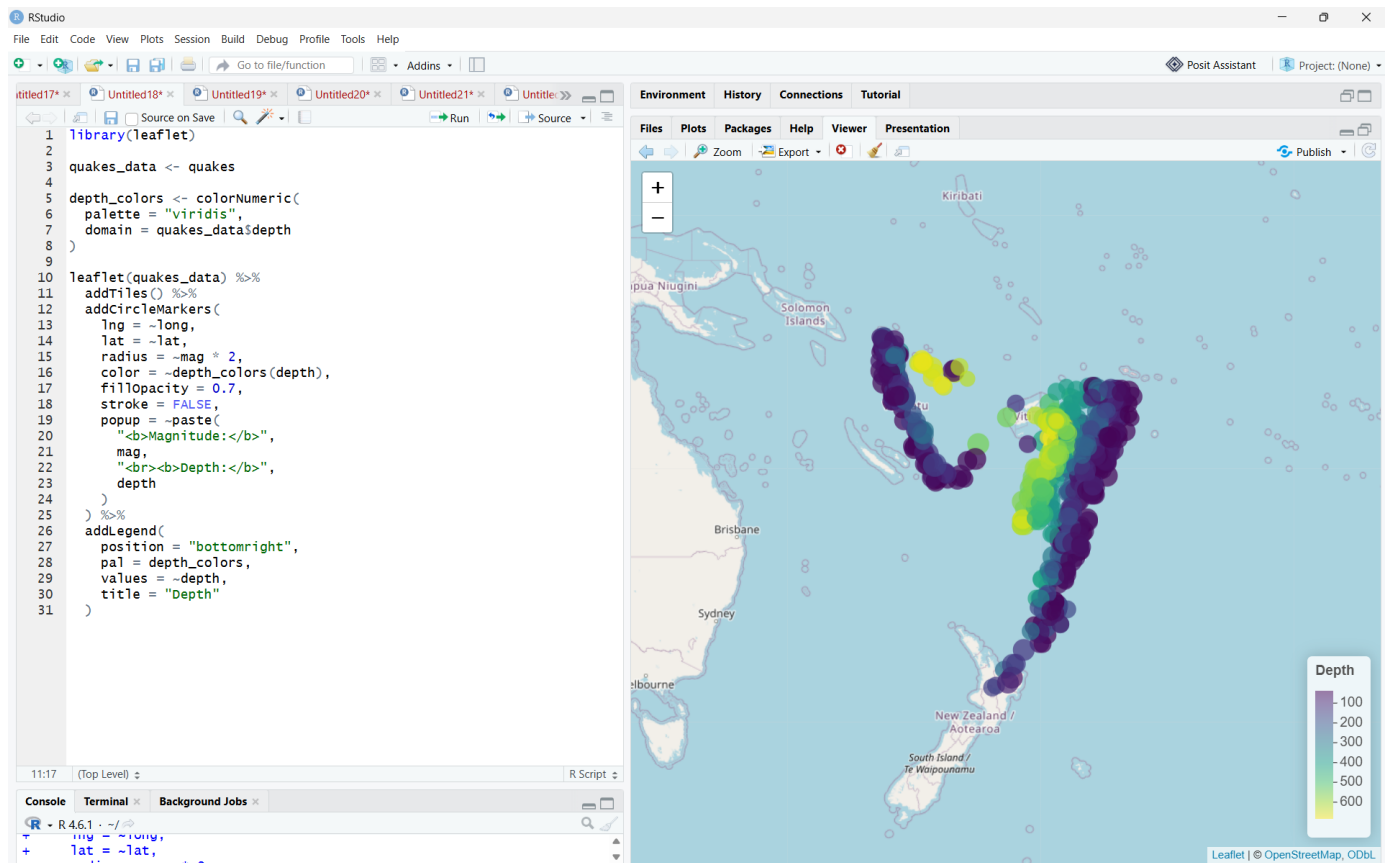
Language: R

Dataset provided: quakes.csv

Using the `leaflet` package, plot the earthquake locations as circle markers on an interactive map, sizing each marker by `mag` and coloring by `depth` (e.g., using `colorNumeric()`). Add popups showing magnitude and depth on click. Briefly describe any spatial clustering of high-magnitude events.

Skills tested: `geom_polygon()/geom_sf()` choropleths, sequential color scales, `leaflet::addCircleMarkers()`, `colorNumeric()`, interactive popups, spatial clustering interpretation.

Answer:



The earthquake points show the spatial distribution of seismic events near Fiji. Larger circles represent higher-magnitude earthquakes. The spatial concentration of larger markers indicates areas where relatively high-magnitude seismic events are clustered.

Question 4: Cartogram with Map Projections

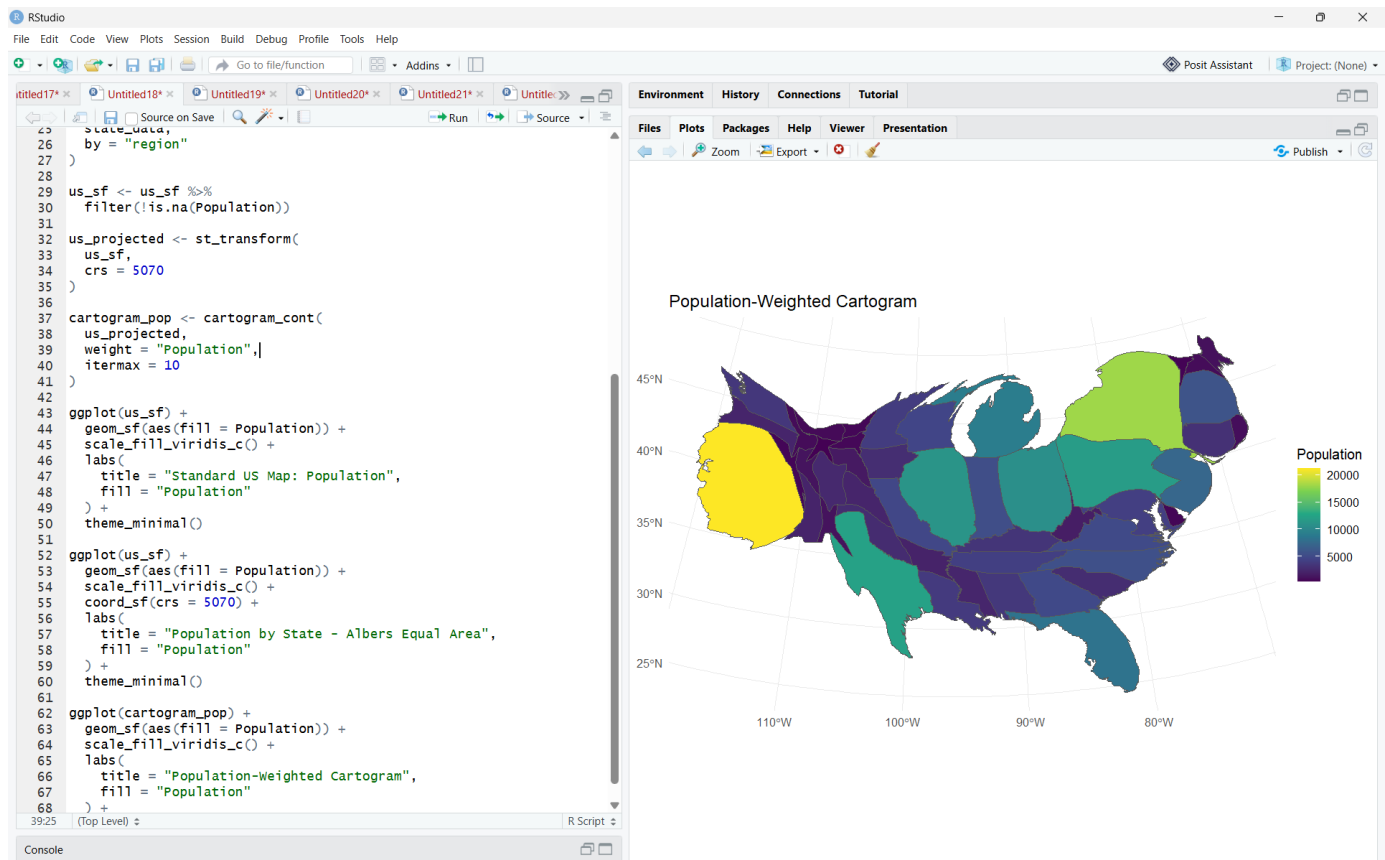
Language: R

Dataset provided: `state_x77.csv` (join to US state boundary polygons, e.g. via the `maps` package)

Join `state_x77.csv` to US state boundary polygons and construct a population-weighted cartogram using the `cartogram` package (`cartogram_cont()` or `cartogram_ncont()`, `sf`-based workflow), distorting each state's area in proportion to its Population. Render the cartogram alongside a standard (undistorted) choropleth of the same Population variable, using two different projections for the standard map (e.g., `coord_sf(crs = ...)` with an Albers equal-area projection vs. a simple longitude/latitude projection). Discuss how the cartogram changes the visual impression of which states appear "large" compared to the standard map, and how the choice of projection affects the standard map's shape.

Skills tested: *sf* workflows, *cartogram_cont()*/*cartogram_ncont()*, *coord_sf()* and projections (CRS), comparing distorted vs. true-area representations.

Answer:



You will obtain:

1. Standard US population map.
2. Population map using an Albers equal-area projection.
3. Population-weighted cartogram.

In the standard map, states retain their true geographic shapes and areas. In the population-weighted cartogram, states with larger populations become visually larger, while states with smaller populations shrink. Therefore, highly populated states have greater visual importance in the cartogram.

The choice of projection also changes the apparent shape and spatial arrangement of states. An Albers equal-area projection is useful for comparing areas while reducing distortion across the continental United States.

Question 5: 3D Geospatial Heatmap

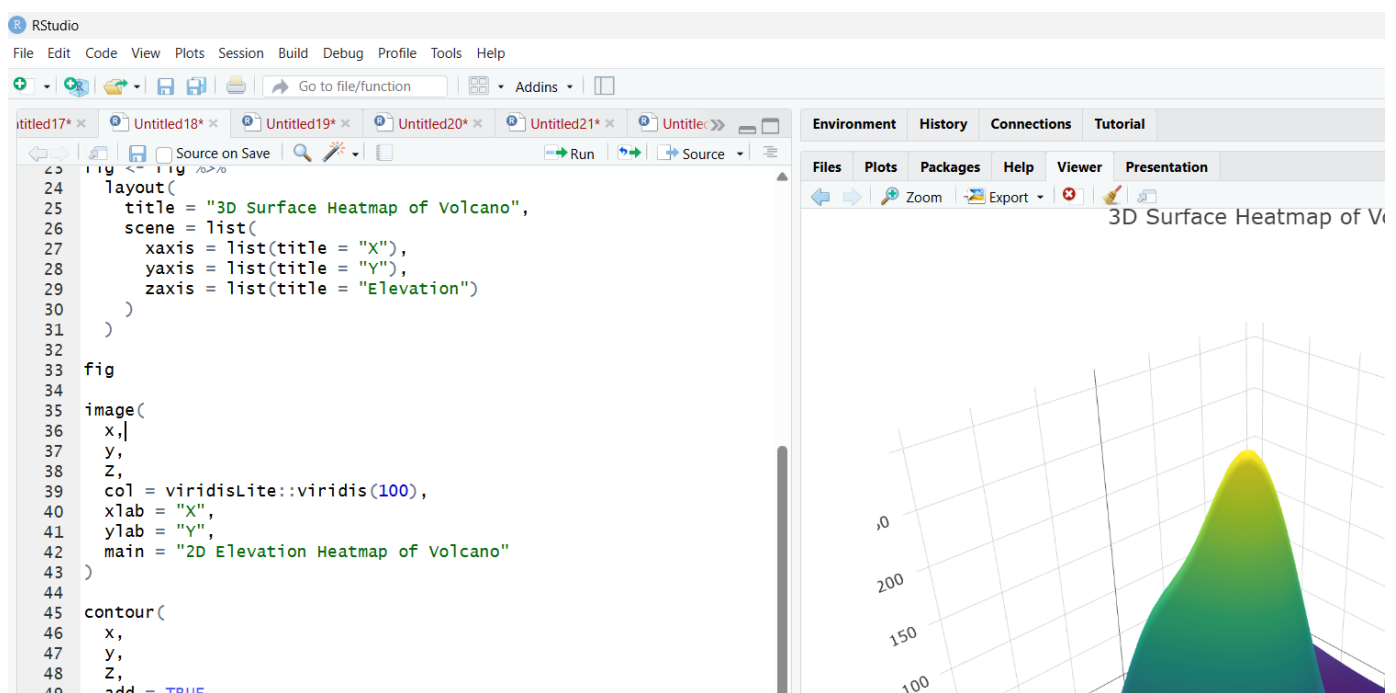
Language: Python

Dataset provided: volcano_elevation.csv

Load volcano_elevation.csv (columns: x, y, elevation — a 61×87 grid representing the Maunga Whau volcano digital elevation model). Pivot the data into a 2D elevation grid and create an interactive 3D surface heatmap using Plotly (plotly.graph_objects.Surface, with a suitable colorscale such as "Earth" or "Viridis" mapped to elevation). Add axis titles and a title, and enable hover tooltips showing elevation at each grid point. Additionally produce a 2D top-down heatmap (e.g., plotly.graph_objects.Heatmap or seaborn.heatmap) of the same data for comparison. In 2–3 sentences, discuss what the 3D surface view reveals about the volcano's shape that the 2D heatmap alone does not (or vice versa).

Skills tested: pandas pivot(), plotly.graph_objects.Surface, 3D colorscales and hover tooltips, 2D heatmap comparison, matplotlib/seaborn or plotly for 2D view.

Answer:



The Plotly graph displays an interactive 3D volcano surface.

You can:

- Rotate the volcano.
- Zoom in/out.
- Hover over points.
- View the exact elevation.
- Observe peaks and valleys.